SECTION–I(i) : (Maximum Marks : 57)

- This section contains **NINTEEN** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : –1 In all other cases

1. $Na[BH_4]$ is formed by reaction of :-
(A) Two electron deficient hydrides
(B) Two saline hydrides
(C) One saline and one electron deficient hydride
(D) One metallic and one electron deficinet hydride

2. Choose correct statement about various water softening method :
(A) In Colgon's method sodium hexametaphosphate ($Na_6P_6O_{18}$) is used
(B) In permutit method sodium aluminium silicate is used
(C) In synthetic resin method, all soluble minerals from water are removed.
(D) All of these

3. Choose correct order :
(A) Solubility $Be(OH)_2 < Ba(OH)_2$ (B) Thermal stability $LiOH < CsOH$
(C) Magnetic moment $O_2 > O_2^-$ (D) All are correct

4. Which of the following is true about water :
(A) Hardness can be removed by passing through ion–exchange resin.
(B) Its presence can be detected by anhydrous $CuSO_4$
(C) It has maximum density at 4°C
(D) All of these

5. Out of Cobalt, Scandium, Iron and Zinc simple salts, which is attraced in a magnetic field?
(A) Zinc simple salts only (B) Cobalt simple salts only
(C) Both Cobalt and Iron simple salts (D) Both Scandium and Zinc simple salts

6. Among the VO_4^{3-}, CrO_4^{2-}, MnO_4^{2-} & FeO_4^{2-}, which of the following pair of oxoanion is strongest oxidant & weakest oxidant?
(A) VO_4^{3-}, CrO_4^{2-} (B) FeO_4^{2-}, VO_4^{3-} (C) MnO_4^{2-}, FeO_4^{2-} (D) CrO_4^{2-}, MnO_4^{2-}

7. Fire extinguisher contain H_2SO_4 and :
(A) $BaCO_3$ (B) $NaHCO_3$ solution (C) Na_2CO_3 (D) $CaCO_3$

8. When sulphur is heated with $NaOH_{(aq.)}$ the compounds formed are:
(A) $Na_2S + H_2O$ (B) $Na_2SO_3 + H_2O$
(C) $Na_2S + Na_2S_2O_3 + H_2O$ (D) $Na_2S_2O_3 + H_2O$

9. Among the following which has maximum solubility in water?
(A) KOH (B) CsOH (C) LiOH (D) RbOH

10. When KCl is heated with conc. H_2SO_4 and solid $K_2Cr_2O_7$, we get :
(A) Chromyl chloride (B) Chromous chloride
(C) Chromic chloride (D) Chromic oxide

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11. Which sequence of reaction show chemical relation between sodium and its compound?

(A) $Na + O_2 \longrightarrow Na_2O \xrightarrow{HCl_{(aq.)}} NaCl \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(B) $Na + O_2 \longrightarrow Na_2O \xrightarrow{H_2O} NaCl \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(C) $Na + H_2O \longrightarrow NaOH \xrightarrow{HCl} NaCl \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{\Delta} Na$

(D) $Na + H_2O \longrightarrow NaOH \xrightarrow{CO_2} Na_2CO_3 \xrightarrow{HCl} \underset{\text{Molten}}{NaCl} \xrightarrow{\text{electrolysis}} Na$

12. For alkali metals, which one of the following trends is incorrect?
(A) Hydration energy : $Li^+ > Na^+ > K^+ > Rb^+$ (B) Ionization energy : Li > Na > K > Rb
(C) Density : Li < Na < K < Rb (D) Atomic size : Li < Na < K < Rb

13. Hydrogen peroxide is now generally prepared on industrial scale by the :
(A) Action of H_2SO_4 on barium peroxide
(B) Action of H_2SO_4 on sodium peroxide
(C) Electrolysis of 50% H_2SO_4 & followed by hydrolysis
(D) Burning hydrogen in excess of oxygen

14. When silicon is boiled with caustic soda solution, the gas evolved is :
(A) O_2 (B) SiH_4 (C) H_2 (D) None of these

15. Which of the following reactions does not form gaseous product?
(A) $PbO_2 + H_2O_2 \longrightarrow$ (B) $PbS + H_2O_2 \longrightarrow$
(C) Acidified $KMnO_4 + H_2O_2 \longrightarrow$ (D) $Cl_2 + H_2O_2 \longrightarrow$

16. The value of .....is less for D_2O as compared to H_2O.
(A) density (g mL^{-1}) at 20°C (B) Boiling point
(C) Dielectric constant at 20°C (D) Latent heat of vaporisation at 20°C

17. The tendency of 3d-metal ions to form stable complexes is due to their :
(A) Very low ionization energies (B) Variable oxidation state
(C) Strong electronegative state (D) High charge/size ratio and vacant d-orbital

18. The maximum number of unpaired electrons are in :
(A) Fe^{2+} (B) Fe^{3+} (C) Fe (D) Fe^{4+}

19. The correct order of decreasing radius is :-
(A) La > Ce > Eu > Yb (B) La > Eu > Ce > Yb
(C) Yb > Eu > La > Ce (D) Eu > La > Ce > Yb

SECTION-I(ii) : (Maximum Marks: 4)

- This section contains **ONE** question.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks	: +4	If only (all) the correct option(s) is (are) chosen.
Partial Marks	: +3	If all the four options are correct but ONLY three options are chosen.
Partial Marks	: +2	If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks	: +1	If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks	: 0	If none of the options is chosen (i.e. the question is unanswered).
Negative Marks	: –2	In all other cases.

20. Which of the following facts are the evidence for ionic nature of LiH.
(A) Molten LiH (m.p. 691°C) conducts electricity
(B) H_2 is liberated at the anode on electrolysis of the molten LiH.
(C) When LiH reacts with water liberates H_2
(D) Li gets deposited at cathod on electrolysis of molten LiH.

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